Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/15/2007
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Project Title:	Structure & Function of Neurovestibular Adaptation & Re-adaptation in the Utricle Following Extended Periods of Hypergravity Exposure		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-04
Start Date:	10/01/2004	**End Date:**	09/30/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2009 per A. Chu/ARC (7/08) Funding reinstated thru 9/30/08, per J. Cohen (HQ) 8/06		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kondrachuk, Alexander Ph.D. (Dept. Theoretical Physics, Institute of Physics, Natl. Acad. Sci., Kiev, Ukraine)		
Grant/Contract No.:	ARC Intramural PI		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The overall goal of the proposed research is to identify the structural and functional changes that occur in the utricle of the vestibular labyrinth as the result of transitions from one gravitational level into another. This information is critical for advancing our understanding of neurovestibular adaptation and its sequelae. Our strategy is to use centrifugation as a tool for temporarily altering the gravitational stimulus. We are particularly interested in determining the afferent firing rate and response characteristics and the morphological changes in the hair cell-afferent complexes found in the utricle that are modified when the organism moves from normal gravity (1G) to hyper-gravity (hyper-G) and in the opposite sense.
In the utricular otolith organ it was found that the afferent response sensitivity to head acceleration (13, fish) as well as the number of synaptic bodies in hair cells (56, rats) increase as the result of exposure to microgravity (μG). A preliminary result showed that exposure to hyper-G by centrifugation decreases the number of synaptic bodies (56, rats). The period of afferent re-adaptation to normal gravity in fish (13) was found to be close to the period of behavioral adaptation in humans (55). Re-adaptation is defined here as the process of adaptation to normal gravity. These results suggest the possibility that the process of adaptation to μG might be mimicked by the process of re-adaptation from a hyper-G level. However, the experiments that would combine the results of μG and hyper-G experiments for the same species are absent. Further, the correlations of the synaptic re-arrangement with afferent sensitivity following exposure to altered gravity for the same species remain undefined. The purpose of the present proposal is to fill these gaps.

Our model is the adult toadfish, Opsanus tau. Fish provide an excellent model system for gravitational research (7). We have everything in place to successfully complete the proposed set of experiments and modeling. Of prime importance is the fact that in the same species a comprehensive evaluation of the neural response to μG (13) and hyper-G (present proposal) exposure will be made. There are four specific aims: Aim 1. Response characteristics of utricular afferents to Centrifugation. Fish will be placed in groups and exposed to different magnitudes (1.5 and 3 g) and duration (1-16 days) of centrifugation. On Center Controls and ground controls are included. Using electrophysiological techniques, individual utricular afferents will be physiologically characterized based on gravitoinertial acceleration testing. Afferent response properties to efferent vestibular activation will be specified. The duration of afferent recording session is tempered by the need to freeze the tissue in time to permit the measurements of structure-function correlations. To match the conditions under which post-flight experiments were performed (13) changes in afferent sensitivity caused by hyper-G will be studied from 1-24 hrs after centrifugation, and re-adaptation after 24-96 hrs in 1G . Aim 2. Morphology of hair cell-afferent synaptic complexes. Combining the techniques of labeling of individual afferents with light and transmission electron microscopy a complete characterization of the afferent is possible: its physiology, its dendritic morphology in the macula, and the extent and quantification of synapses formed from the innervated hair cells. Visualization of complex 3-D microcircuitry will also reveal precise connectivity within the macula that is essential for understanding neurovestibular adaptation. Synaptic body counts will be performed to address the fundamental question of the correlation of afferent response sensitivity with hair cell synapse number. Aim 3. Merging the structure and function of otolith afferents and their relationships to hyper-G. The specific times and thresholds of the afferent response adaptation and re-adaptation and the structural changes associated with the hair cell-afferent synapse will be determined by varying the magnitude and duration of centrifugation. Aim 4. Modeling. Based on known methodologies and the present experimental results, research will be directed towards the establishment of phenomenological models describing the adaptive changes in utricular afferents in altered gravity conditions. We will examine the relationship between the afferent response and the number of synapses in normal and in hyper-G conditions. A model of toadfish otolith mechanics in these conditions will be also developed. The four specific aims are directed to achieve the main goals: 1) to test the correlations of change in the gravitational level and the change in afferent response characteristics caused by head acceleration and the potential underlying changes in the morphology of the associated hair cell-afferent synaptic complexes, 2) to examine the role of efferent vestibular system on these correlations, 3) to test (propose) the possible mechanisms responsible for these correlations; and 4) to test if the transition from hyper-G to normal gravity can be used to mimic the effects of neural adaptation to μG. We expect that convergence of findings in μG and hyper-G will provide a model of neurovestibular plasticity, and will lead to the development of novel test protocols for both ground- and space-based research. As we learn about the basic cellular processes in vestibular neurons, countermeasures will eventually be developed to reduce the symptoms associated with Space Motion Sickness or even modify synaptic sensitivity in preparation of entry into a higher gravity level.

The overarching hypotheses to be evaluated are: One, the gravitoinertial sensitivity of otolith nerve afferents is altered as a consequence of the animal's exposure to hyper-G conditions as well as μG. It is reasoned that the transition from 1G to hyper-G will decrease otolith afferent sensitivity (output in impulses per sec/input in G), and the transition from hyper-G back to 1G will increase it (and back to normal levels). Thus, the transition from hyper-G to normal gravity may serve as an imitator of neurovestibular sequelae to μG. Two, hair cell-afferent synaptic complexes are modified by the altered gravity condition: the up- or down regulation of synaptic bodies is specific to the direction of transition from 1G. The change in number of synapses is directly correlated with afferent response sensitivity. Three, afferent adaptation mechanisms may involve either self-learning processes confined to the hair cell or responses of the hair cell-afferent system to feedback signal(s) that originate from a misbalance of information coming from otolith and extra-otolith contributor(s), or a combination of both. (These mechanisms will be considered by modeling based on the experimental findings.) Four, efferent vestibular actions are not uniform on hair cells and afferents, and it is proposed that the changes brought on in otolith afferents by the transitions from one gravity state to another will be most prevalent and coincident in afferents strongly affected by efferent activation. If this proves to be the case, then the next step is to directly query the efferent neurons themselves.

The overall goal of the proposed research is to identify the structural and functional changes that occur in the utricle of the vertebrate vestibular labyrinth as the result of transitions from one gravitational level into another. This information is critical for advancing our understanding of neurovestibular adaptation and its sequelae, and for the effective design and validation of countermeasures to benefit astronaut health and performance. Our strategy is to use centrifugation as a tool for temporarily altering the gravitational stimulus. We are particularly interested in determining the afferent firing rate and response characteristics and the morphological changes in the hair cell-afferent complexes found in the utricle that are modified when the organism moves from normal gravity (1G) to hyper-gravity (hyper-G) and in the opposite sense. This study is a natural analog to our previous studies aboard STS-90 and 95, and will provide a comprehensive evaluation of the neural response to μG and hyper-G in the same species.

We expect that convergence of findings in μG and hyper-G will provide a solid model of neurovestibular plasticity, and will lead to the development of novel test protocols for both ground- and space-based research. As we learn about the basic cellular processes in vestibular neurons, countermeasures will eventually be developed to reduce the symptoms associated with Space Motion Sickness, benefit sensorimotor coordination and performance, or even down regulate

synaptic sensitivity in preparation of entry into a higher gravity level.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The functional importance of head stabilization is seen by our ability to change gaze orientation and maintain binocular fixation of objects of interest during not only standing but also during locomotor activities, and our ability to detect self-motion and adaptively react to applied perturbations of the head and body. In the space environment the vestibular system is profoundly affected by the loss of he gravity vector. However, conditions on Earth can seriously affect the individual's ability to functional normally. In addition aging places a serious challenge to the vestibular and neural centers responsible for posture and movement control. Disorders of the vestibular system affect head, gaze and posture stabilization, and can dramatically alter the patients willingness to induce self-motion and can lead to trepidation of normal activities. These studies will also provide important insights into vestibular function, and may help in the diagnosis and treatment of vestibular patients seen in the clinic.

We have made significant progress towards reaching our stated goals. Four separate and successful centrifugation trails were conducted to examine the response of utricular nerve afferents to 3g and 1.5g hyper-gravity stimulation and the time course of re-adaptation to Earth's 1g following centrifugation. The centrifugation stimulus and configuration have worked perfectly and without incident. Further, we have been able to successfully record the electrophysiological responses of more that 60-70 separate afferents in each experiment, twice what we had anticipated. This increase in sample has provided a very reliable measure to evaluate response between experiments.
We have completed the following experiments. Four (4) fish were studied following 3G exposure at 1, 2, 3, and 8 days; five (5) fish at 4 and 16 days; and three (3) fish at 5 days (total 26 fish). The 3-Day and 5-Day groups were added to our protocol based on the results we obtained at 4-Day exposure to 3G (this will be detailed below). For the Re-adaptation experiments, three (3) fish were studied following 4-days of 3G exposure and re-adaptation periods of 1, 2, 4, 6 and 8 days (total 15 fish). Eight laboratory controls have been completed. In the next group of Re-adaptation experiments, three (3) fish were studied following 16-days of 3G exposure and re-adaptation periods of 2, 6 and 8 days, and four (4) at 4-days of re-adaptation to 1G (total 13 fish). In the latest centrifugation trial, we successfully completed four (4) fish exposed to 1.5G at 4-days and 16-days (total 8 fish). The next centrifugation will complete the experiments needed for the Re-adaptation portion of the study (approximately 15 fish) and the 1.5G exposure and Re-adaptation (approximately 15 fish). In the next period On-Center control experiments also will be conducted. To this end we recently acquired a separate and adequately equipped room that is perfect for these experiments. An 80 ft-lb DC torque motor/rate table was installed in the room. We are currently developing the suitable aquarium for this task.

Previously, we studied the response of utricular afferents following exposure to microgravity (Boyle et al. [2001] Neural readaptation to 1G following return from space. J. Neurophysiol. 86: 2118-2122). Data were collected from 2 orbital shuttle missions: STS-90 and STS-95. Within the first day after landing, the magnitude of response to an applied acceleration was significantly greater, 3-fold upregulation of the reponse sensitivity, than for controls. Within about 30 hours post-landing the response return to normal values. The number of synaptic bodies in type II hair cells in rodent is labile, increasing following exposure to microgravity (Ross 1993, 1994, 2000), suggesting an increase in synaptic density was responsible for the upregulation of sensitivity. The results obtained in the present study are not complete, but nonetheless exciting. The response sensitivity begins to increase after 1, 2 and 3 days of 3-g exposure, and by 4-days the sensitivity is 3-fold ($p<0.0001$) increased, actually mimicking the microgravity data. The sensitivity then begins to fall, and by 16-days of 3-g exposure the sensitivity is 2-fold ($p<0.0001$) downregulated. This is an important finding: the response of the utricle to hypergravity is modifiable and bi-modal. As mentioned in the experimental groups above, re-adaptation was studied. Here, the subjects are centrifuged at 4- or 16-days, and returned to their habitat (1-g environment), and studied at variable durations. With 4 days of recovery (Re-adaptation) following 3-g exposure for 4- and 16-Days the gain did not return (fall or rise) to control levels. The data, however, must be viewed still as preliminary because i) because of the fact that we can successfully record from more than twice the anticipated number of afferents, detailed analyzes of each experiment requires considerable amount of time, and therefore not all data obtained have been analyzed, ii) additional centrifugation experiments are ongoing and scheduled, and iii) on-center controls have not yet been performed.

Matching the function with the structure requires analysis of the synaptic structures in the hair cells. Synaptic body counts are ongoing. We examine the number of synaptic bodies in hair cells within the striolar and extra-striolar regions of the utricular macula. Synaptic bodies are determined using transmission electron microscopy from serial sections at 5K-40K maginification. To date we have data analyzed from one control utricle and one utricle from a 16-day exposure to 3-g. For 16-day exposure to 3-g there was a reduction in synaptic bodies in both the striolar, more than 2-fold decrease, and extra-striolar regions, and this was highly significant with the striolar region ($p<0.0001$). This finding matches the reduction in response sensitivity recorded in this experiment. This is an extremely labor-intensive effort. We had a set back in this effort: the CCD camera had a malfunction and the camera was damaged beyond repair during shipping. As result we were required to purchase a new CCD system which resulted in delays in manufacturing and installation. We are now back in full progress.

Functional Considerations and General Conclusions:

A change in gravity level has a profound effect on how an organism maintains equilibrium, and the neural response might involve the peripheral otolith receptors, the brain or both. In turn, the change of the sensitivity of peripheral otolith receptors in altered G may depend on the change of physical parameters of the otolith structures responsible for transferring the dynamics of head into the dynamics of the hair cell bundle deformations, as well as on the change of properties of local peripheral neural structures, first of all, synapses.

I. Let us analyze the first option: the change of physical parameters of fish otolith structure responsible for the sensitivity of afferent in altered gravity, as it is formulated in (1): imp/sec*g. According to (1), the sensitivity of toadfish utricular afferents measured within 10 hr of the shuttle landing (17 and 9 days flights) was sevenfold greater than the control mean. We will assume that the response of otolith afferent is proportional to the displacement of hair cell bundles. These displacements correspond to the displacement of the gel layer in which these hair cells are embedded. For rough estimate of the coefficient of proportionality, K, we can use the relationship between the gel layer displacement, U, and acceleration, A, of the head in the case of two-layered model of otolith represented by gel layer and otolith (stone) itself (2). (The analysis of various models of otolith structures and function is presented in (3)):

Ux(z)=[(1+sg)hg2/Eg][2((Delta roho/hg)+Delta rg)(z/hg)-Delta rg(z/hg)2)]Ax;

(note: this format here does not support equation text and symbols) where Ux(z) denotes the component of displacements of the gel layer parallel to the macular surface at the distance z from this surface. Ax is component of head acceleration parallel to the macular surface. Delta ro and Delta rg are the differences of densities of the otolith and the endolymph and the gel layer and endolymph correspondingly, ho and hg are thicknesses of otolith and gel layer, Eg is Young's modulus, and sg is Poisson's ratio of the GL layer. Thus Ux=K Ax. This equation shows that the gel displacement could be considerably increased only if Young's modulus of gel is decreased. The variation of other parameters of cannot give us the essential growth of K. Since we assumed the linear dependence of afferent response and hair bundle displacements, it means that after micro-G Eg has to be seven times less than in normal 1G. Although mechanical parameters of biogels are very sensitive to the changes of ion compositions and electrical fields, such great changes of gel Young's modulus seem to be unrealistic. It should be noted that the value of Eg has to be balanced by two contradictory criteria (4): it should be high enough to prevent the otolith from dropping on receptor cells and be low enough to provide the sufficient displacement of this cells caused by head acceleration. It means that if Eg decreases essentially in micro-G, the stone will be lying on the macular surface after return to 1G conditions. Therefore the change of utricular afferent sensitivity in altered G may be connected with neural plasticity of otolith organ.

II. One of the hypotheses considered in the project was formulated as follows: "afferent adaptation mechanisms (unquote, to altered gravity, quote) may involve either self-learning processes confined to the hair cell or responses of the hair cell-afferent system to feedback signal(s) that originate from a misbalance of information coming from otolith and extra-otolith contributor(s), or a combination of both".

It may be assumed that the possible "goal" of such self-regulation or adaptation is to compensate the change of the parameters of the system caused by the external influence and to return them into the normal physiological range of the analyzing system (5). The mechanism of adaptation depends on the mechanism of feedback existing in this system. Because of this, the goals of our work (5,6) were the structuring and analysis of the hypothesis of feedback and the demonstration of the ambiguity of possible effects of the altered gravity on formation and function of gravi-receptors derived from the data from mollusks and fish.

Summarizing these data, the feedback hypothesis may be formulated as a set of the following basic assumptions (Model) (5,6): 1) The gravi-sensing system development consists of a set of successive time-dependent parts (subprograms of development) acting during specific periods. 2) There are some mechanisms of feedbacks of local and neural origins to control the course of subprogram fulfillment and the transition from one program to another. The local feedbacks are confined by small volume of precocious receptors. 3) If the afferents and efferents are still immature, the "neural" feedback cannot work and any effect of the change of gravity on gravi-receptors during this period has local origin. It is possible that different feedbacks are involved at different periods in the development of the gravi-sensing system in altered gravity: at the earlier stages they are local and in the mature state they may have neural-like efferent nature (7, 8). 4) The altered gravity may cause direct effects in developing gravireceptors that are not related to any mechanisms designed to compensate for the possible changes of environment normally experienced by the organism. 5) Altered gravity influences those mechanisms of the system whose parameters and processes depend on gravity level. These may be internal parameters of subprograms or control parameters that determine transition from one program to another. It is possible that several feedbacks corresponding to different stages of gravi-receptor development would be sensitive to altered gravity. The examples are the changes of statolith and statoconia in Aplysia in hypergravity conditions observed in in vitro experiments (9,10). 6) It has to be stressed that every response of the gravi-sensing system to the change of altered gravity in the course of gravi-receptor formation may be realized ONLY within the limits of the existing subprograms designed during animal evolution in normal gravity conditions, possibly, for absolutely different purposes. It means, for instance, that lack of otolith weight in _G may be interpreted by the organism as a shortage of mineral component in otolith composition. To remove this limitation, probably, needs the participation of neural feedback. Returning to the previous question about the possibility of sevenfold decrease of Eg it should be noted that it is difficult to imagine the situation in 1 G conditions that would need such crucial changes of gel properties.

(It was supposed that "otolith mineralization hitherto is the only biomineralization process known, which is neuronally regulated in adaptation to an environmental parameter (i.e., gravity)"(11). Our opinion is that the possible neural regulation of otolith growth cannot be considered as some kind of adaptation to the change of gravity as an environmental parameter, because the gravity was never changed during animal development, but this neural feedback may be the response to the effect of the altered gravity on some physical, chemical or informational processes participating in otolith formation).

III. The experimental findings (1, 13) indicate that sensory plasticity exists in the utricle following exposure to altered gravity conditions. However the mechanisms of neural adaptation in these experiments are still not clear.

Two different Models may be considered as a basis of possible explanation. The first Model is relied on self-learning synaptic contacts whose properties are changing in accordance with the change of pre- and post-synaptic activities (14, 15). This mechanism of adaptation is supposed to be confined in pre-and post-synaptic relationship and does not need any external reference signal. It may be called 'local feedback"(5). If we apply this model to utricular "receptor-afferent" system, it will mean that mechanical stimulus being transformed in hair cells in altered gravity will carry some information on the gravity level that initiates synapse modification including the synaptogenesis.

According to the analysis of feedback mechanisms (5), we may expect that the local feedback is more specific for developing organism with immature nervous system.

The second Model supposes the inclusion of a Reference system and corresponding modulating signal in the "receptor-afferent" chain. It suggests that there are two useful signals in this chain: output afferent signal A that carries the information about the cell response to mechanical stimulation and feedback signal B, the function of which is to modify the state of receptor in accordance with the first signal A information. This hypothesis says that the change of otolith dynamics caused by altered gravity generates signal B as a result of comparison of the signal A with some external reference signal C, where it is reasonable to assume that this signal C is coming from the system that is not influenced by gravity.

Task Progress:

Possible candidates for such system can be identified through investigation of mechanisms that allow species to distinguish between gravitational and inertial accelerations. There are indications (16, 17) that the brain resolves this problem by combining and comparing the signals from multiple sensors, the semicircular canal signals being the main extra-otolith contributors. Since gravitational information sensed by the otolith is mainly related to the tilts of the head,

this signal is accompanied by the signal C of the corresponding semicircular canal. When the gravity is altered, the dynamics of the otolith, for instance, the specific time of its reaction to the tilt, will be changed while, as it is supposed, the reaction C of the semicircular canals does not depend on the gravity level. The resulting misbalance induces the feedback signal B directed to the receptors to restore the balance, possibly, by modifying their sensitivity. Assuming this hypothesis is correct, the μG and hyper-G conditions will correspond to different situations. We may expect that in μG the relatively slow tilt will not cause the displacement of the otolith relative to receptor cells and thus the signal A will be absent while the signal C will exist. Meanwhile in hyper-G the signal B will be not only present but enhanced relative to 1G conditions.

The functioning of such feedback will be effective if the Reference system would not depend on the signal A information. In our case it suggests the independence of the function of the system of semicircular canals on gravity. The works (18-21) are devoted to the analysis of this problem.

The function of the semicircular canals (SC) as hydromechanical sensors of angular head rotation is based on the equality of specific gravities of the cupula and endolymph. Without this equality, the information provided by the SC would depend on the spatial orientation of canals relative to gravity and the axis of rotation. However, we may assume that the equality of densities of the cupula and endolymph is not precise but approximate because: i) the compositions of the cupula and endolymph are different; ii) these densities can change due to the change of pressure, temperature, metabolic processes or due to the ingestion of proper doses of substances that can pass a blood-labyrinth barrier and has a specific gravity different from endolymph (e.g. ethanol, heavy water, glycerol). Hence, we may expect that due to the very high sensitivity of the SC (~0.058 sec-2), even the small density differences (DD) (~10-4g/cm3) can influence its function, and that this influence depends on the conditions of canal stimulation. The works (18-21) aimed to examine this hypothesis and analyze the parameters of the SC and mechanical stimulation under which the effect of the difference of densities on the functioning of a cupulo-endolympatic system could be observed. The consideration was based on a simplified mathematical model where the canal was approximated by a torus and the cupula by a piston. If the densities are equal, the model transforms to Steinhauzen's phenomenological equation.(The various approaches to modeling the SC are described in (22)). The present analysis is focused on two cases of canal stimulation: the rotation with constant angular velocity w and harmonic oscillation of the angular velocity; and two types of spatial orientation of the SC: i) the gravity vector and axis of rotation lie in the plane of the canal, ii) the axis of rotation and gravity vector are normal to the canal plane. The difference of densities creates the moments of inertial force relative to the axis of the canal and thus cupula dynamics becomes dependent on the orientation of the gravity vector relative to the canal plane and the axis of rotation, as well as the distance between the axis of rotation and the center of semicircular canal (l). It leads to the shift of the cupula to a new position of equilibrium that depends on the gravity and parameters of rotation l and w. The estimates show that this shift can exceed the threshold of sensitivity. A new position of cupula equilibrium may invoke the illusion of the permanent angular acceleration of the head even when the frequency =const. Due to the DD the cupula elasticity coefficient in the equation of the motion becomes effectively dependent on the parameters of harmonic stimulation that results in the onset of cupula oscillations with multiple frequencies and the distortion of cupula dynamics relative to harmonic stimulation. Even for a very small DD, its effect can be revealed by the appropriate conditions of stimulation. Different effects can take place in the different canals of the same labyrinth at the same time. The effects caused by the DD can be developed due to the ingestion of some substances and/or in conditions of artificial gravitation on space stations.

It is important that in altered gravity there are at least two conditions that may cause or enhance the DD effects: the variation of intra-labyrinthine pressure and the change of parameters of heat exchange in inner ear.

These results show that the functioning of the system of semicircular canals has to be considered as gravity-sensitive. Therefore the alteration of gravity level may influence not only the functioning the gravireceptors, but the state of the Reference system too and thus distort the signal B corresponding to misbalance of the receptor signal A and reference signal C. This fact once again indicates that alteration of gravity can not be considered as an usual variation of environmental parameters for which the mechanisms of neural adaptation are designed. Thus the adaptation to such stimulus may have non-traditional form.

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Conference Presentations for this Project to Date:

1) Kondrachuk A, Boyle R (2005) The role of possible feedback mechanisms in the effects of altered gravity on formation and function of gravireceptors of mollusks and fish. Joint Life Science Conference, July 2005, Cologne, Germany, p. 41-42. (Abstract)

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